

## Introduction to CSS Grid

### What is CSS Grid?

CSS Grid is a two-dimensional layout system.

It controls:

- Rows
- Columns
- Gaps
- Alignment

Grid example

```
[ Product 1 ][ Product 2 ][ Product 3 ][ Product 4 ]  
[ Product 5 ][ Product 6 ][ Product 7 ][ Product 8 ]
```

Description:

A table also has rows and columns.

But CSS Grid is:

- Flexible
- Responsive
- Easy to modify
- Used in modern web development

### Grid Terminology

#### Grid Container

The parent element.

```
.products {  
  display: grid;  
}
```

#### Grid Item

Each child element.

```
<section class="products">  
  
  <article>Product 1</article>  
  <article>Product 2</article>  
  <article>Product 3</article>  
  
</section>
```

### Grid Columns

```
.products {
  display: grid;
  grid-template-columns: repeat(4, 1fr);
}
```

Description:

- repeat(4,1fr) means 4 equal columns.
- 1fr means one fraction of available space.

### Connect CSS File

create two files:

- index.html
- style.css

Modify HTML.

```
<!DOCTYPE html>
<html>
<head>
  <title>KLECommerce</title>

  <link rel="stylesheet" href="style.css">
</head>

<body>

</body>
</html>
```

### Style Header and Navigation

In style.css write:

```
body{
  margin:0;
  font-family:Arial;
  background:#f4f4f4;
}

header{
  background:#0D47A1;
  color:white;

  display:grid;
  grid-template-columns:1fr 2fr 1fr;

  align-items:center;
  padding:15px;
```

```

}

nav{
  background:#1976D2;
  color:white;

  display:grid;
  grid-template-columns:repeat(8,1fr);

  text-align:center;
  padding:12px;
}

```

## Header Grid

grid-template-columns:1fr 2fr 1fr

Means:

| Column | Purpose      |
|--------|--------------|
| 1fr    | Logo         |
| 2fr    | Search       |
| 1fr    | Login / Cart |

|      |        |            |
|------|--------|------------|
| Logo | Search | Login/Cart |
|------|--------|------------|

## Convert Products into Grid

### Step 1: Modify HTML

Change the product section.

```

<section class="products-section">

<h2>FEATURED PRODUCTS</h2>

<div class="products">

<article class="product">
  
  <h3>Men's Shirt</h3>
  <h2>₹999</h2>
  <button>Add to Cart</button>
</article>

<article class="product">

```

```


<h3>Rice Bag</h3>
<h2>₹1200</h2>
<button>Add to Cart</button>
</article>

<article class="product">
  
  <h3>Smart Phone</h3>
  <h2>₹15000</h2>
  <button>Add to Cart</button>
</article>

<article class="product">
  
  <h3>Washing Machine</h3>
  <h2>₹22000</h2>
  <button>Add to Cart</button>
</article>

</div>

</section>

```

## Step 2: Add CSS

```

.products-section{
  padding:20px;
}

.products-section h2{
  background:#1565C0;
  color:white;
  text-align:center;
  padding:15px;
}

.products{
  display:grid;

  grid-template-columns:repeat(4,1fr);

  gap:20px;

  margin-top:20px;
}

.product{
  background:white;

```

```
border:1px solid #ddd;

text-align:center;

padding:15px;
}

.product button{
  background:#1565C0;
  color:white;
  border:none;
  padding:10px 20px;
}
```

## Convert Services into Grid

Modify HTML.

```
<section class="services">

<div>Easy Returns</div>

<div>24 × 7 Support</div>

<div>Secure Payment</div>

<div>Fast Delivery</div>

</section>
```

Add CSS.

```
.services{
  display:grid;

  grid-template-columns:repeat(4,1fr);

  background:#E8F5E9;

  text-align:center;

  padding:20px;

  gap:15px;
}

.services div{
  background:white;
```

```
padding:15px;

border:1px solid #ccc;
}
```

## Responsive Grid

Modern websites must work on mobile phones.

Replace:

```
grid-template-columns:repeat(4,1fr);
```

with

```
grid-template-columns:repeat(auto-fit,minmax(200px,1fr));
```

Description:

- 200px = minimum width
- 1fr = flexible width
- auto-fit = automatically adjusts number of columns

This is the big advantage over tables.

## Complete CSS Grid Code

**style.css**

```
body{
  margin:0;
  font-family:Arial;
  background:#f4f4f4;
}

header{
  background:#0D47A1;
  color:white;
  display:grid;
  grid-template-columns:1fr 2fr 1fr;
  align-items:center;
  padding:15px;
}

nav{
  background:#1976D2;
  color:white;
  display:grid;
  grid-template-columns:repeat(8,1fr);
  text-align:center;
  padding:12px;
}
```

```
.products-section{
  padding:20px;
}

.products-section h2{
  background:#1565C0;
  color:white;
  text-align:center;
  padding:15px;
}

.products{
  display:grid;
  grid-template-columns:repeat(auto-fit,minmax(200px,1fr));
  gap:20px;
  margin-top:20px;
}

.product{
  background:white;
  border:1px solid #ddd;
  text-align:center;
  padding:15px;
}

.product button{
  background:#1565C0;
  color:white;
  border:none;
  padding:10px 20px;
}

.offers{
  background:#FFF176;
  padding:20px;
  text-align:center;
}

.services{
  display:grid;
  grid-template-columns:repeat(auto-fit,minmax(180px,1fr));
  background:#E8F5E9;
  text-align:center;
  padding:20px;
  gap:15px;
}
```

```
.services div{  
  background:white;  
  padding:15px;  
  border:1px solid #ccc;  
}
```

```
footer{  
  background:#263238;  
  color:white;  
  text-align:center;  
  padding:20px;  
}
```